

Screenshots

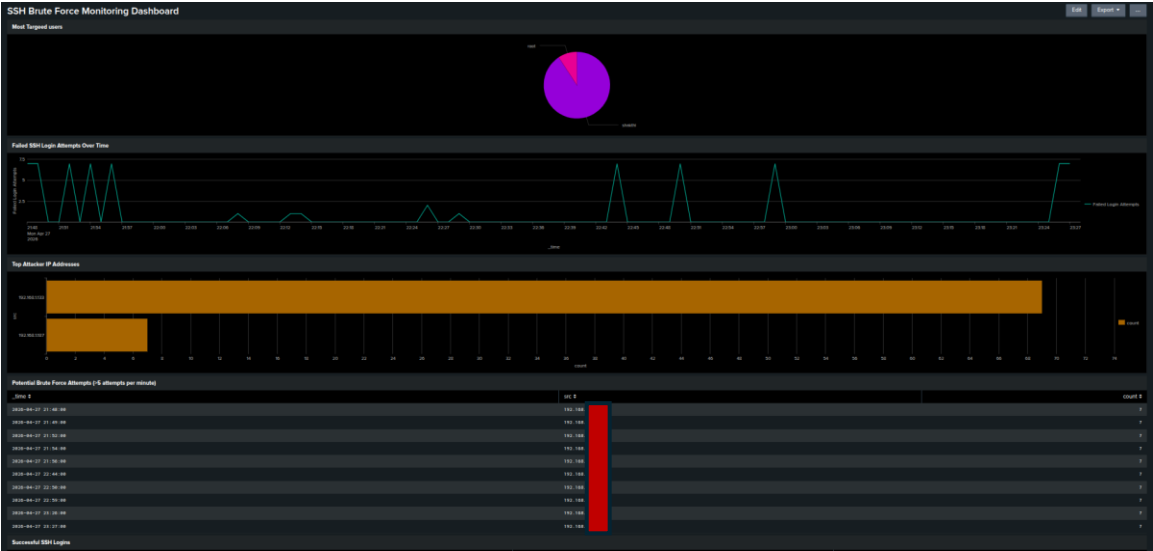


Figure A.8: Dashboard Summary Statistics Panel

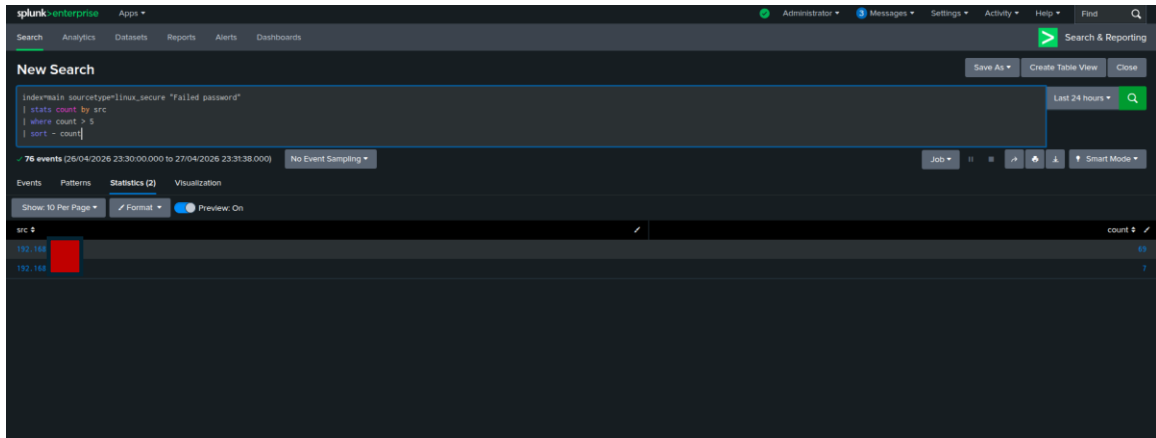
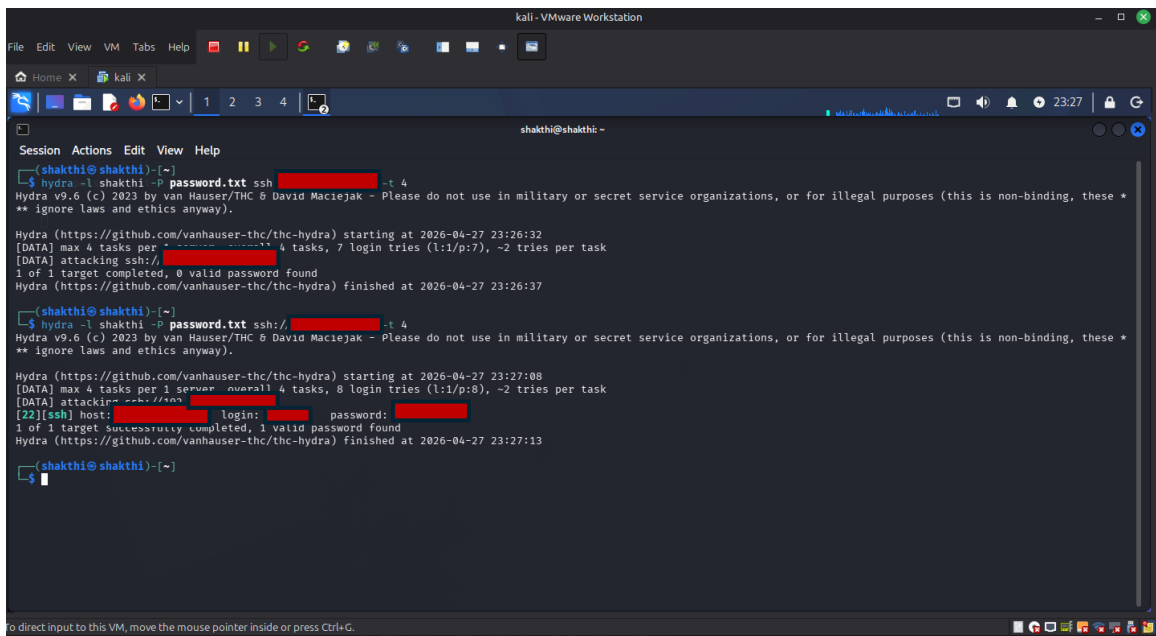


Figure A.6: Brute Force Detection Query Results

```
shakthi@shakthi-ThinkPad-T480: ~  
shakthi@shakthi-ThinkPad-T480: ~$ sudo cat /opt/splunkforwarder/etc/system/local/inputs.conf  
[monitor:///tmp/*]  
disabled = false  
sourcetype = malware  
index = malware  
  
[monitor:///var/log/auth.log]  
index = main  
sourcetype = linux_secure  
disabled = false  
  
[monitor:///var/log/syslog]  
index = main  
sourcetype = syslog  
disabled = false  
  
[monitor:///var/log/kern.log]  
index = main  
sourcetype = syslog  
disabled = false  
  
[monitor:///var/log/ufw.log]  
index = main  
sourcetype = ufw  
disabled = false  
shakthi@shakthi-ThinkPad-T480: ~$
```

"Never give up on your d
you know what you

Figure A.3: inputs.conf Configuration



```
kali - VMware Workstation
File Edit View VM Tabs Help
Home X kali X
shakthi@shakthi ~
Session Actions Edit View Help
shakthi@shakthi ~
$ hydra -l shakthi -P password.txt ssh [redacted] -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *
** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-27 23:26:32
[DATA] max 4 tasks per [redacted] 4 tasks, 7 login tries (l:1/p:7), ~2 tries per task
[DATA] attacking ssh://[redacted]
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-27 23:26:37

shakthi@shakthi ~
$ hydra -l shakthi -P password.txt ssh://[redacted] -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *
** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-27 23:27:08
[DATA] max 4 tasks per 1 server overall 4 tasks, 8 login tries (l:1/p:8), ~2 tries per task
[DATA] attacking ssh://[redacted]
[22][ssh] host: [redacted] login: [redacted] password: [redacted]
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-27 23:27:13

shakthi@shakthi ~
$
```

Figure A.4: Hydra Attack Execution

splunk enterprise Apps Administrator Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search

index=* sourcetype="linux_secure" Save As Create Table View Close

286 events (26/04/2026 22:30:00.000 to 27/04/2026 23:25:30.000) No Event Sampling Last 24 hours Job 1 2 3 4 5 6 7 8 ... Smart Mode

Events (286) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection Deconstruct 1 hour per column

Format Show 20 Per Page View List < Prev 1 2 3 4 5 6 7 8 ... Next >

SELECTED FIELDS

- host: 1
- pid: 12
- process: 5
- source: 1
- sourcetype: 1
- src_ip: 2

INTERESTING FIELDS

- action: 5
- app: 1
- date_hour: 3
- date_index: 1
- date_minute: 25
- date_second: 1
- date_second: 30
- date_second: 30
- date_year: 1
- date_year: 1
- eventtype: 5
- index: 1
- interval: 1
- name: 4
- punct: 26
- reason: 2

Time	Event
27/04/2026 23:20:55.634	2026-04-27T23:20:55.634817+05:30 shakthi-ThinkPad-T480 polkitid[594]: Unregistered Authentication Agent for unix-process:14020:912288 (system bus name :1.171, object path /org/freedesktop/PolicyKit1/AuthenticationAgent, locale C) host = shakthi-ThinkPad-T480 pid = 694 process = polkitd source = /var/log/auth.log sourcetype = linux_secure
27/04/2026 23:20:55.628	2026-04-27T23:20:55.628650+05:30 shakthi-ThinkPad-T480 polkitid[594]: Registered Authentication Agent for unix-process:14020:912288 (system bus name :1.171 [flatpak list], object path /org/freedesktop/PolicyKit1/AuthenticationAgent, locale C) host = shakthi-ThinkPad-T480 pid = 694 process = polkitd source = /var/log/auth.log sourcetype = linux_secure
27/04/2026 23:20:49.317	2026-04-27T23:20:49.317813+05:30 shakthi-ThinkPad-T480 polkitid[594]: Unregistered Authentication Agent for unix-process:13863:911621 (system bus name :1.178, object path /org/freedesktop/PolicyKit1/AuthenticationAgent, locale C) host = shakthi-ThinkPad-T480 pid = 694 process = polkitd source = /var/log/auth.log sourcetype = linux_secure
27/04/2026 23:20:49.226	2026-04-27T23:20:49.226841+05:30 shakthi-ThinkPad-T480 polkitid[594]: Registered Authentication Agent for unix-process:13863:911621 (system bus name :1.178 [flatpak list], object path /org/freedesktop/PolicyKit1/AuthenticationAgent, locale C) host = shakthi-ThinkPad-T480 pid = 694 process = polkitd source = /var/log/auth.log sourcetype = linux_secure
27/04/2026 23:17:01.171	2026-04-27T23:17:01.171546+05:30 shakthi-ThinkPad-T480 CRON[13646]: pam_unix(cron:session): session closed for user root host = shakthi-ThinkPad-T480 pid = 13646 process = CRON source = /var/log/auth.log sourcetype = linux_secure
27/04/2026 23:17:01.164	2026-04-27T23:17:01.16418+05:30 shakthi-ThinkPad-T480 CRON[13646]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0) host = shakthi-ThinkPad-T480 pid = 13646 process = CRON source = /var/log/auth.log sourcetype = linux_secure
27/04/2026 23:11:13.528	2026-04-27T23:11:13.528886+05:30 shakthi-ThinkPad-T480 sudo: pam_unix(sudo:session): session closed for user root host = shakthi-ThinkPad-T480 process = sudo source = /var/log/auth.log sourcetype = linux_secure
27/04/2026 23:11:12.249	2026-04-27T23:11:12.24953+05:30 shakthi-ThinkPad-T480 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by shakthi(uid=1898) host = shakthi-ThinkPad-T480 process = sudo source = /var/log/auth.log sourcetype = linux_secure

Figure A.5: Authentication Logs in Splunk Search

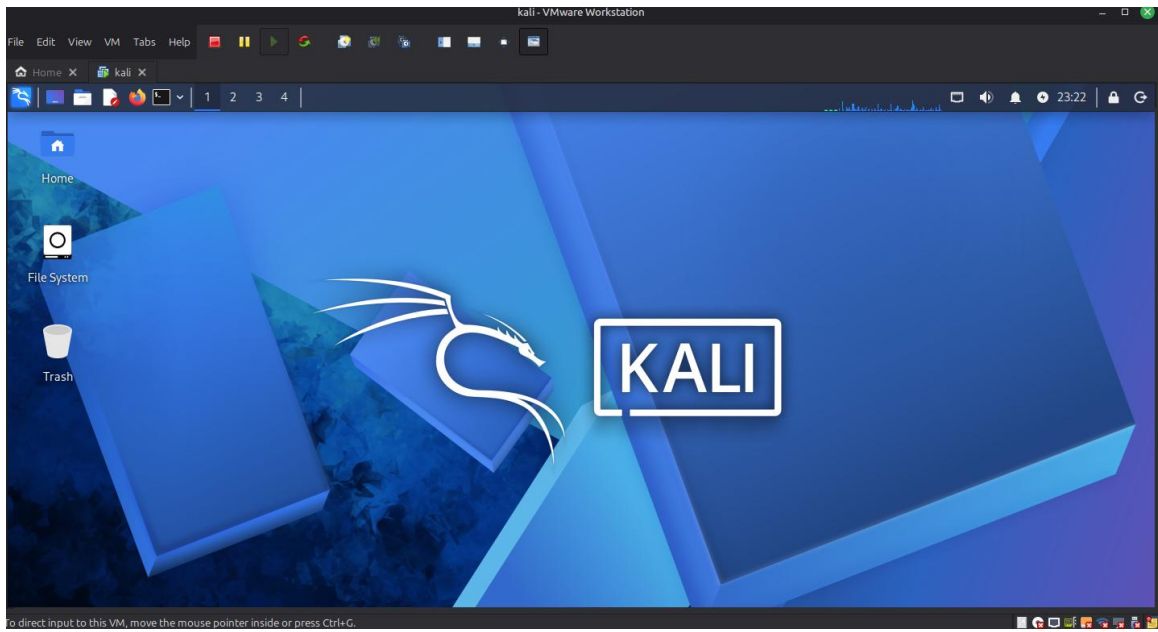


Figure A.1: Attacker Machine (Kali Linux) Setup

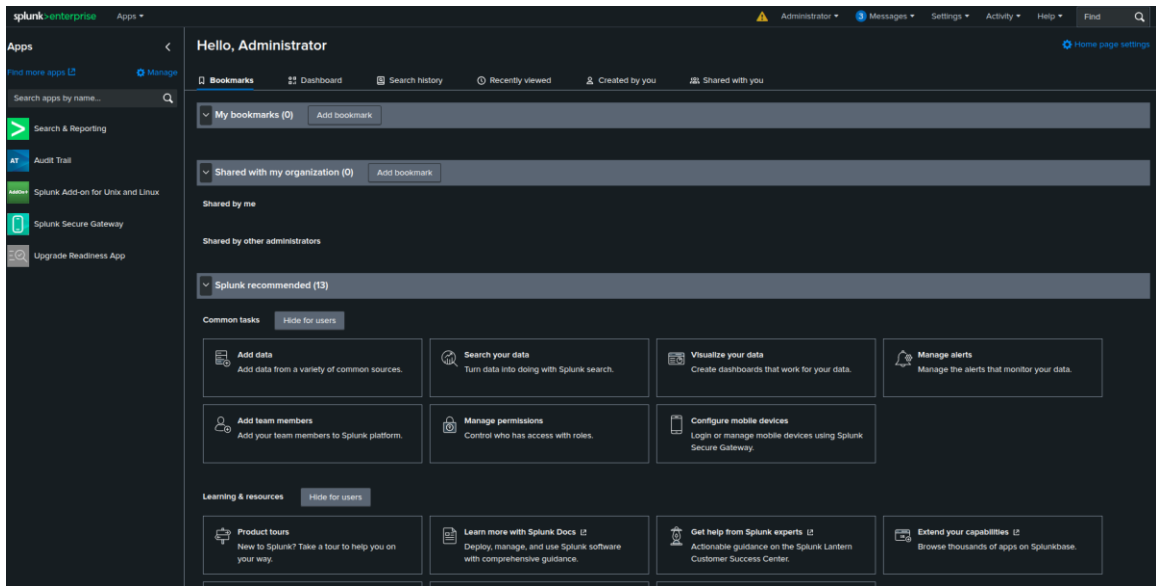


Figure A.7: Splunk Dashboard – Complete View

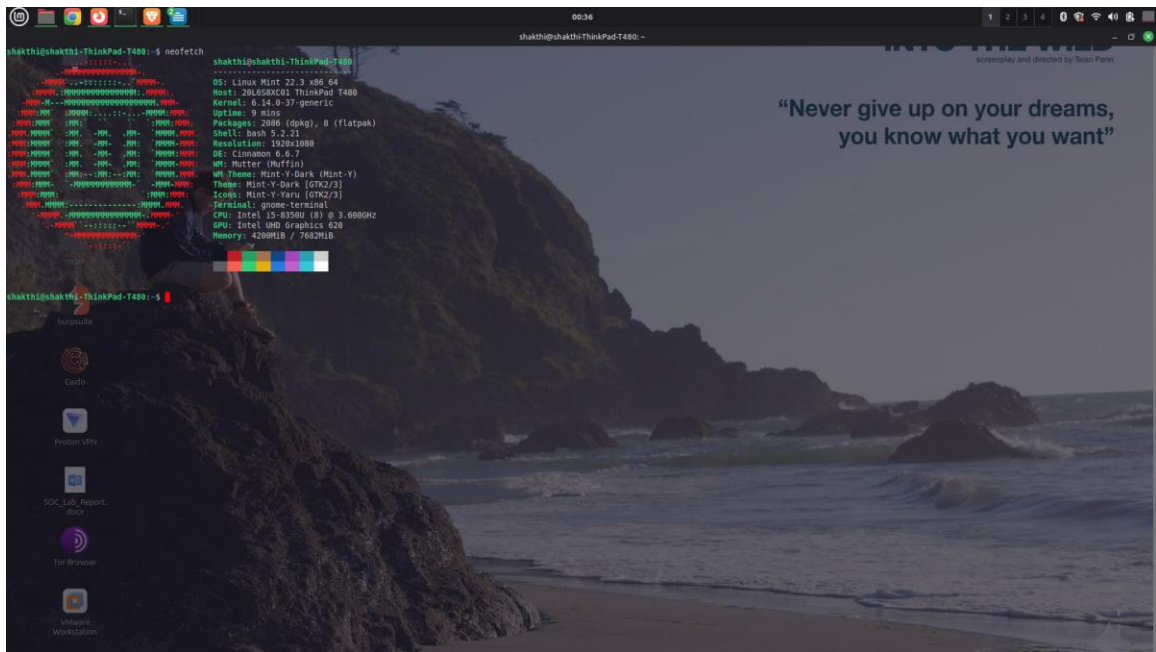


Figure A.2: Target Machine (Linux Mint) Setup